MILITARY SPECIFICATION

HOSE, SYNTHETIC RUBBER, SYNTHETIC FIBER REINFORCED, AND END FITTINGS, REUSABLE, FOR FLEXIBLE HOSE ASSEMBLIES (HIGH PRESSURE SIZES -4 THROUGH -32)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers synthetic fiber reinforced rubber hoses and fittings in sizes -4 through -32 for use in high pressure applications.

1.2 Classification. End fittings for hose covered by this specification shall be of the following types and nominal sizes:

Type E - 90 degree dogleg elbow, hose sizes -6 to -16 (see figure 6).
Type G - SAE, 37 degree tube swivel to hose, straight, hose sizes -4 to -32 (see figure 1).
Type H - Gasket seal union, male half to hose, straight, tube sizes 1/8 inch to 2 inches (see figure 2).
Type HL - Gasket seal union, male half to hose, 90 degree elbow, tube sizes 1/8 inch to 2 inches (see figure 3).
Type I - Male half O-ring union, straight, hose sizes -6 through -32 (see figure 4).
Type IL - Male half O-ring union, 90 degree elbow, hose sizes -6 through -32 (see figure 5).

The gasket seal union, types H and HL are obsolete and are to be used for replacement purposes only.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications. Unless otherwise specified, the following specifications of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:30Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought.
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

MILITARY

MIL-T-1368 - Tube and Pipe, Nickel-Copper Alloy, Seamless and Welded.
MIL-T-16420 - Tube, Copper-Nickel Alloy, Seamless and Welded, (Copper Alloy Numbers 715 and 706).
MIL-H-24136 - Hose, Synthetic Rubber, Synthetic Fiber Reinforced and End Fittings, Reusable, for Flexible Hose Assemblies, General Specification For.

(Copies of specifications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Hoses and end fittings furnished under this specification shall conform to the requirements of MIL-H-24136 and as specified herein.

3.2 Dimensions and pressures. Hose dimensions and pressures shall be in accordance with table I.

TABLE I. Dash number, performance characteristics and dimensions.

Dash no. (hose size)	Pressure (lb/in^2)		Hose dimensions (inches)				Hose reinforcement		Minimum bend radius (inches)	Impulse peak pressure (lb/in^2)
	Working (max)	Burst (min)	I.D. Min	I.D. Max	O.D. Min	O.D. Max	Construction	No. of layers		
-4	2500	10,000	0.234	0.266	0.844	0.906	Braided	3	4	3750
-6	2000	8,000	0.359	0.39	0.968	1.031	Braided	3	5	3000
-8	2000	8,000	0.465	0.531	1.094	1.156	Braided	3	7	3000
-12	1500	6,000	0.718	0.781	1.469	1.531	Braided	3	9.5	2250
-16	1400	5,600	0.968	1.031	1.750	1.812	Braided	3	11	2100
-20	1000	4,000	1.211	1.289	1.975	2.069	Braided	3	16	1500
-24	1000	4,000	1.461	1.539	2.203	2.297	Braided	3	20	1500
-32	900	3,600	1.961	2.039	2.702	2.828	Braided	3	25	1350

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:30Z
Check the source to verify that this is the current version before use.

3.3 Fitting types and sizes. Fitting types and sizes shall be as shown on figures 1 through 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance provisions shall be in accordance with MIL-H-24136 and as specified herein.

4.2 Sampling for qualification testing.

4.2.1 Hose. Qualification testing of hose shall consist of hose samples as specified in table II. Samples shall be not less than 15 inches in length and shall be fitted with qualified end fittings from at least two previously qualified suppliers (see 4.2.4).

TABLE II. Number of hose samples for qualification testing.

Hose size (dash no.)	Number of hose samples to be provided
-8	4
-32	5

4.2.2 End fittings. Qualification testing of end fittings only, shall consist of 18 end fittings of the same type as specified in table III. End fitting samples shall be installed on 9 qualified hose samples, in the sizes listed in table II, from at least two previously qualified hoses (see 4.2.4).

TABLE III. Number of end fitting samples for qualification testing.

End fitting size (hose dash no.)	Number of end fitting samples to be provided
-8	8
-32	10

4.2.3 Hose and end fittings combined. In the event that a single contractor of both hose and fittings desires to qualify both hose and fittings, or a hose contractor and a fitting contractor desire to qualify hose and fittings in a combined qualification test, the samples shall be provided as specified in table IV.

4.2.4 Sample requirements when there are less than two previously qualified suppliers. Where there is only one qualified source for qualified samples all the previously qualified samples required shall come from that source. Where there are no previously qualified sources, all samples of hose and fittings shall be submitted by the prospective qualifier.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:30Z
Check the source to verify that this is the current version before use.

TABLE IV. Number of samples for combined qualification testing.

One approved source					
Hose size (dash no.)	No. of hose samples to be provided		End fitting (hose dash no.)	No. of end fittings to be provided	
	Qualifier 2/	Approved 1/ source		Qualifier 2/	Approved 1/ source
-8	2	2	-8	4	4
-32	3	2	-32	6	4

Two or more approved sources						
Hose size (dash no.)	Number of samples to be provided by			End fitting size (dash no.)	Number of end fittings to be provided by		
	Qualifier 2/	Approved 1/ source #1	Approved 1/ source #2		Qualifier 2/	Approved 1/ source #1	Approved 1/ source #2
-8	2	1	1	-8	4	2	2
-32	3	1	1	-32	6	2	2

Hose samples shall be not less than 15 inches in length and shall be coupled on one end with the fitting being submitted for qualification and on the other end with previously qualified end fittings.

1/ Approved source - Previously qualified hose and fittings, or hose or fitting.
2/ Qualifier - Hose and fittings, or hose or fitting to be qualified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:30Z
Check the source to verify that this is the current version before use.

4.2.5 Sequence of tests on sample hose assemblies shall be as specified in table V.

TABLE V. Sequence of qualification tests on sample assemblies.

Test sequence	Sample numbers								
	Size (dash no.) -8				Size (dash no.) -32				
	1	2	3	4	5	6	7	8	9
Proof pressure (see 4.6.1)1/	X	X	X	X	X	X	X	X	X
Stability test (see 4.6.2)1/	X	X	X	X	X	X	X	X	X
Aging test (see 4.6.3)1/		X	X	X	X		X	X	X
Hydraulic fluid circulation (see 4.6.4)1/		X					X	X	X
Impulse test (see 4.6.5)1/				X	X				X
Burst test (see 4.6.6)1/	X					X			
Cold temperature flexing (see 4.6.8)1/			X					X	

1/ See MIL-H-24136.

4.2.6 Satisfactory completion of qualification testing of the samples selected in accordance with tables II, III and IV will qualify the manufacturer for all sizes shown in table I.

5. PACKAGING

5.1 Preparation for delivery requirements shall be in accordance with MIL-H-24136.

6. NOTES

6.1 Intended use. Hose and fittings to this specification sheet are intended for use in shipboard or shore to ship piping systems where pressures do not exceed the rated working pressure and temperatures do not exceed minus 40°F or 200°F.

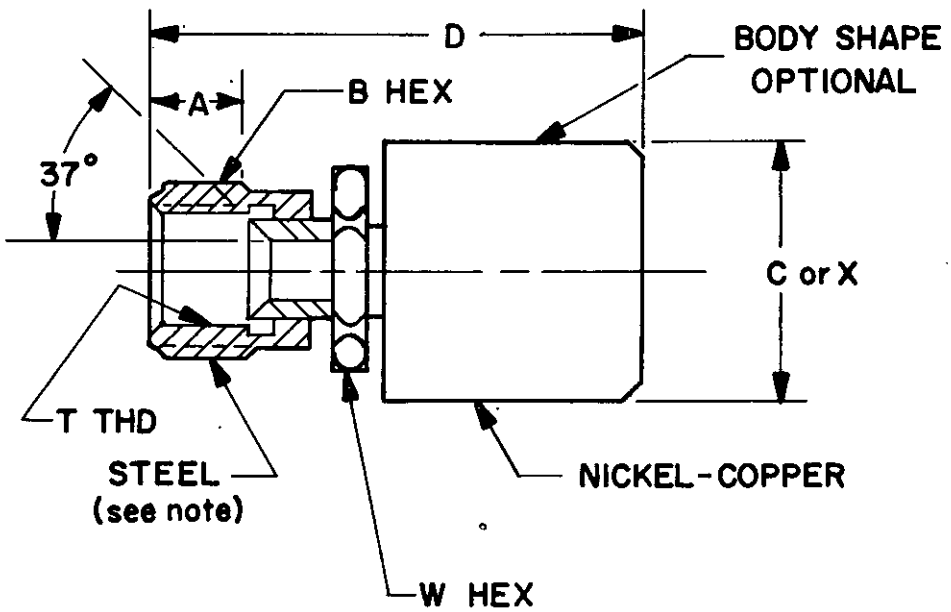
Source: https://assist.dla.mil -- Downloaded: 202610-19T19:30Z
Check the source to verify that this is the currentversion before use.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of end fittings for hoses (see 1.2).

Preparing activity:
Navy - SH
(Project 4730-N874-2)

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:30Z
Check the source to verify that this is the current version before use.



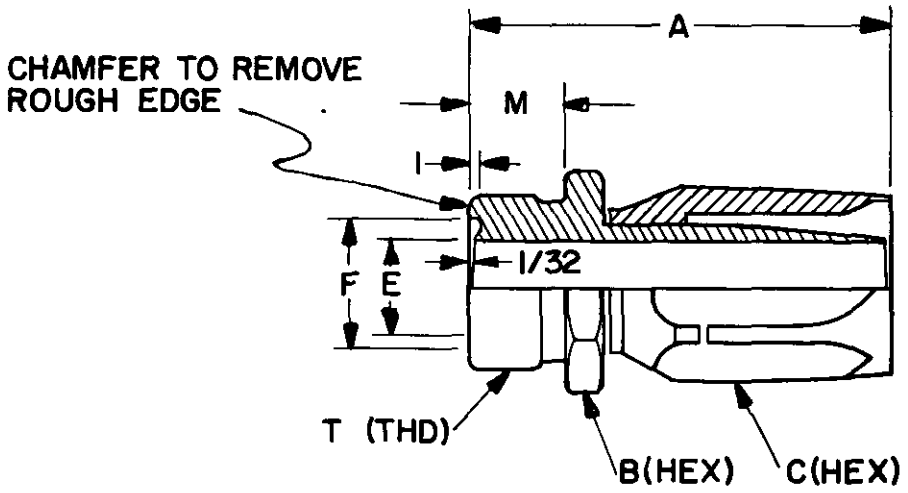
Tube size (nom)	Hose end size (dash no.)	T thread CL-3A	A (max)	B (min)	C (max)	D (max)	X (max)	W Hex (nom)
4	4	7/16-20 UNF	0.37	0.56	1.00	2.75	2.25	0.90
5	4	1/2-20 UNF	0.40	0.62	1.00	2.75	2.25	0.90
6	4	9/16-18 UNF	0.40	0.68	1.38	3.00	2.35	0.90
6	6	9/16-18 UNF	0.40	0.68	1.38	3.25	2.38	0.94
8	6	3/4-16 UNF	0.44	0.87	1.25	3.25	2.38	0.94
8	8	3/4-16 UNF	0.44	0.87	1.50	3.50	3.00	1.00
10	8	7/8-14 UNF	0.52	1.00	1.50	3.50	3.00	1.00
10	10	7/8-14 UNF	0.52	1.00	1.50	3.50	3.00	1.00
12	8	1-1/16-12 UN	0.57	1.25	1.50	3.75	3.00	1.00
12	12	1-1/16-12 UN	0.57	1.25	2.05	4.00	3.50	1.00
14	12	1-3/16-12 UN	0.58	1.37	2.00	4.00	3.50	1.25
16	12	1-3/16-12 UN	0.60	1.50	2.00	4.25	3.50	1.25
16	16	1-5/16-2 UN	0.60	1.50	2.25	4.50	4.25	1.62
20	16	1-5/8-12 UN	0.64	2.00	2.25	4.75	4.25	1.62
20	20	1-5/8-12 UN	0.64	2.00	2.75	5.25	4.50	2.00
24	20	1-7/8-12 UN	0.74	2.25	2.75	5.25	4.50	2.25
24	24	1-7/8-12 UN	0.74	2.25	3.00	5.75	5.00	2.25
32	32	2-1/2-12 UN	0.93	2.87	3.50	6.75	5.75	2.58

SH 12295

NOTES:
1. Dimensions are in inches.
2. Nickel-copper in accordance with QQ-N-281 or nickel-copper-aluminum in accordance with QQ-N-286 is an acceptable substitute for steel.

FIGURE 1. Type G, SAE 37 straight swivel to hose end fitting.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:30Z
Check the source to verify that this is the current version before use.



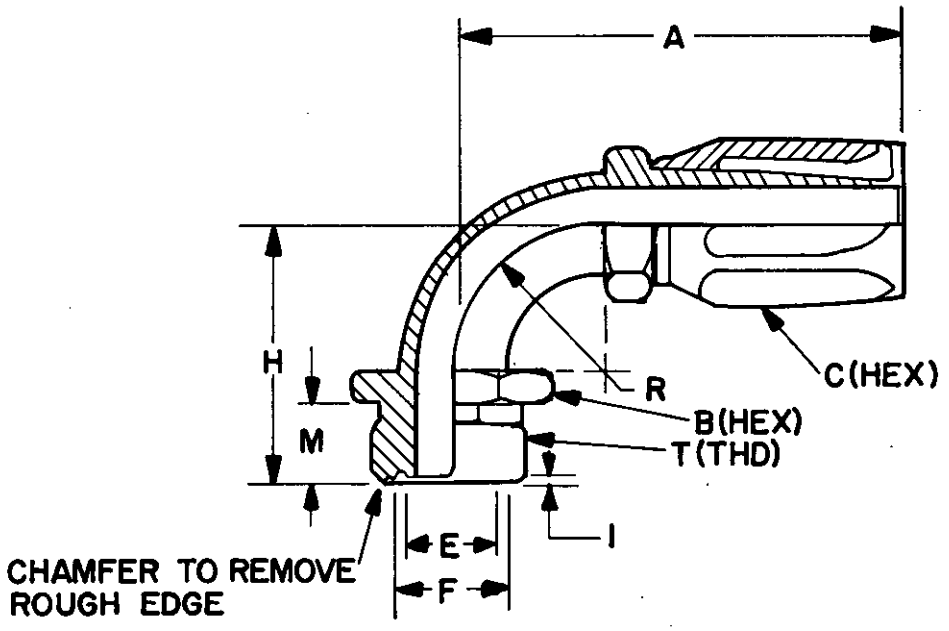
Tube size (nom)	Rubber hose (dash no.)	T thread CL-3	A (max)	B (min)	C (max)	E ±0.005	F ±0.005	I (nom)	M (nom)
1/8	-4	7/8-12 N	3.00	15/16	1.00	0.406	0.594	1/16	0.81
1/4	-6	1-1/8-12 UNF	3.25	1-1/4	1.25	0.406	0.641	1/16	0.88
3/8	-6	1-1/4-12 UNF	3.25	1-3/8	1.25	0.547	0.797	1/16	0.88
3/8	-8	1-1/4-12 UNF	3.50	1-3/8	1.50	0.547	0.797	1/16	0.88
3/8	-12	1-1/4-12 UNF	3.75	1-3/8	1.75	0.547	0.797	1/16	0.88
1/2	-12	1-1/2-12 UNF	4.00	1-5/8	1.75	0.672	1.000	1/16	0.94
1/2	-16	1-1/2-12 UNF	4.25	1-5/8	2.25	0.672	1.000	1/16	0.94
3/4	-16	1-3/4-12 UN	4.50	1-7/8	2.25	0.875	1.188	5/64	1.00
1	-16	2-1/8-12 UN	4.75	2-1/4	2.25	1.094	1.375	5/64	1.12
1	-20	2-1/8-12 UN	5.25	2-1/4	2.75	1.094	1.375	5/64	1.12
1-1/4	-16	2-5/8-12 UN	5.00	2-3/4	2.25	1.344	1.750	3/32	1.25
1-1/4	-20	2-5/8-12 UN	5.50	2-3/4	2.75	1.344	1.750	3/32	1.25
1-1/4	-24	2-5/8-12 UN	5.25	2-3/4	3.00	1.344	1.750	3/32	1.25
1-1/2	-32	2-7/8-12 UN	6.00	3	3.50	1.656	2.094	7/64	1.31
2	-32	3-5/8-12 UN	8.00	3-3/4	3.50	1.938	2.375	1/8	1.38

SH 12296

NOTES:
1. Dimensions are in inches.
2. Fitting assembly material shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, class A, form 1, form 2 condition cold drawn or hot finished or class B.
3. Socket (body) alternative materials shall conform to QQ-N-288, composition B.
4. This fitting is for replacement purposes only and is not to be used in new construction.

FIGURE 2. Type H, gasket seal union, male half to hose, straight.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:30Z
Check the source to verify that this is the current version before use.



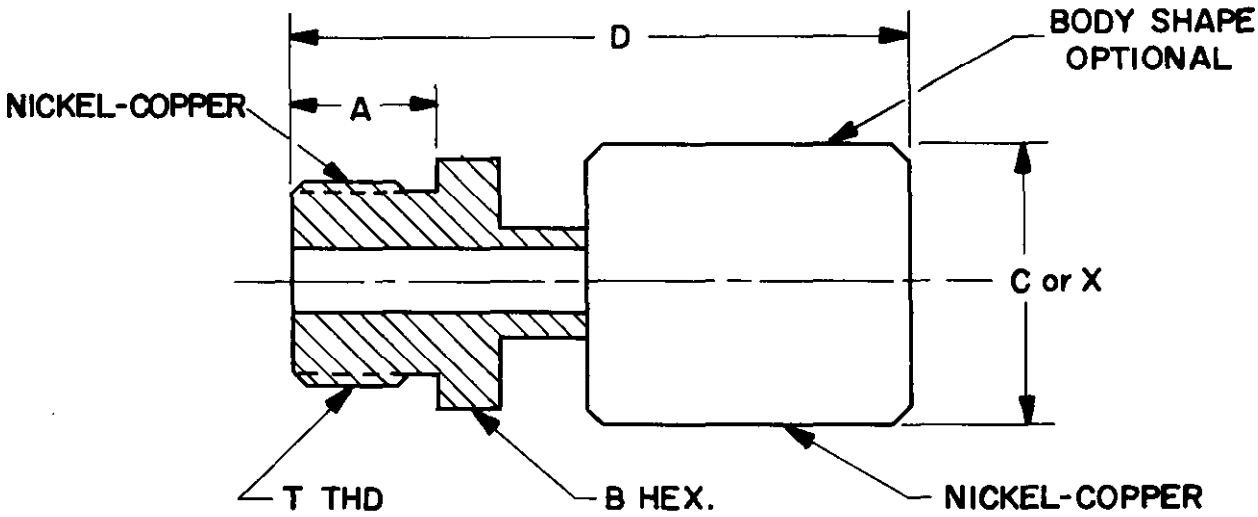
Tube size (nom)	Rubber hose (dash no.)	T thread CL-3	A (max)	B (min)	C (max)	H (max)	E ±0.005	F ±0.005	I (nom)	M (nom)	R (min)
1/8	-4	7/8-12 N	3.55	15/16	1.00	2.00	0.406	0.534	1/16	0.81	0.38
1/4	-6	1-1/8-12 UNF	3.55	1-1/4	1.25	2.25	0.406	0.641	1/16	0.88	0.50
3/8	-8	1/14-12 UNF	3.81	1-3/8	1.50	2.50	0.547	0.797	1/16	0.88	0.75
1/2	-12	1-1/2-12 UNF	4.75	1-5/8	1.75	3.25	0.672	1.000	1/16	0.94	1.25
3/4	-16	1-3/4-12 UN	5.75	1-7/8	2.25	3.75	0.875	1.188	5/64	1.00	1.50
1	-20	2-1/8-12 UN	6.75	2-1/4	2.75	4.25	1.094	1.375	5/64	1.12	1.75
1-1/4	-24	2-5/8-12 UN	7.00	2-3/4	3.00	5.00	1.344	1.750	3/32	1.25	2.00
1-1/2	-32	2-7/8-12 UN	8.00	3	3.50	5.75	1.656	2.094	7/64	1.31	2.50
2	-32	3-5/8-12 UN	8.25	3-3/4	3.50	6.00	1.938	2.375	1/8	1.38	2.50

SH 12297

NOTES:
1. Dimensions are in inches.
2. For hose sizes -4 through -12, the fitting assembly material shall conform to QQ-N-281, class A or B, form 1, condition cold drawn or QQ-N-286, class A, form 1, Ro class B, form 2, condition cold drawn or hot finish.
3. For hose sizes -16 through -32, the fitting assembly material shall conform to MIL-T-16420, composition 70-30, grade II, type 1, class 6000.
4. Socket (body) alternate material shall conform to QQ-N-288, composition B.
5. The 90 degree tube elbow, for fitting sizes 1/8 inch through 3/4 inch shall conform to MIL-T-1368, type 1, condition 1: for fitting sizes 1 through 2 inches to MIL-T-16420, composition 70-30 grade 11, type 1, class 700 special or class 1650 special.
6. This fitting is for replacement purposes only and is not to be used in new construction.

FIGURE 3. Type HL, gasket seal union, male half to hose, 90 degree elbow.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:30Z
Check the source to verify that this is the current version before use.



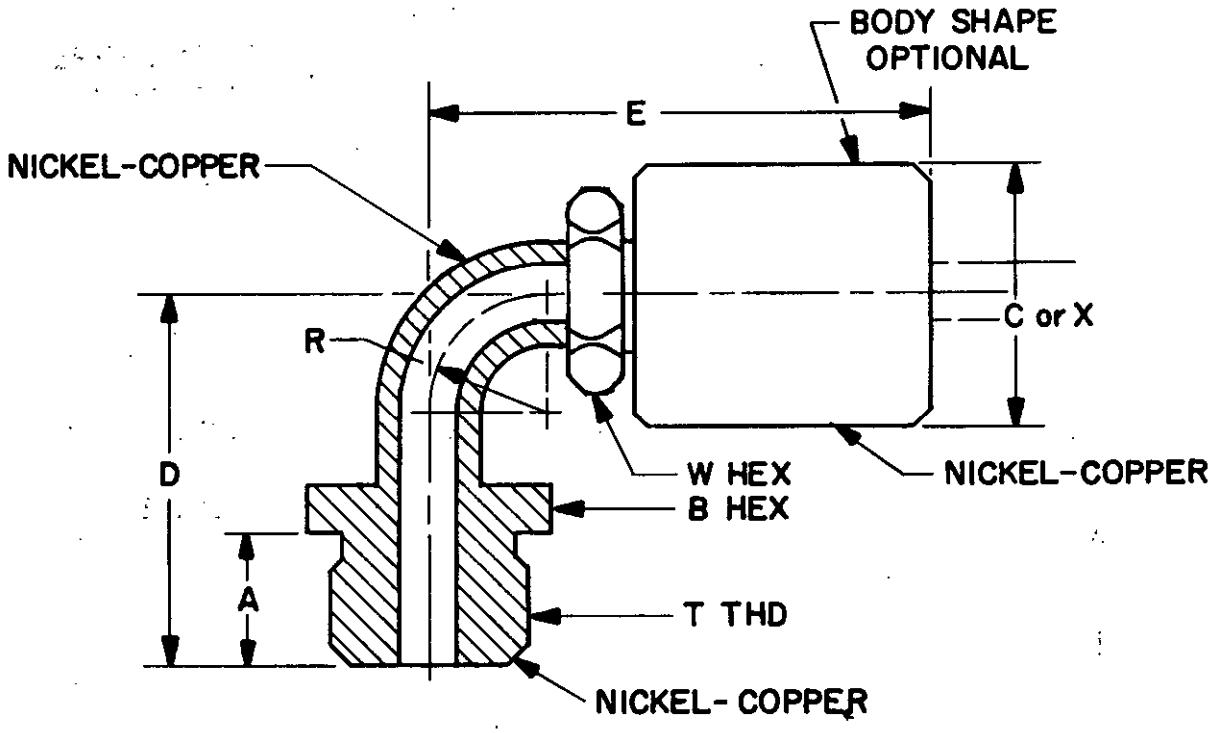
Tube size (nom)	Hose end size (dash no.)	T Thread CL-3A	A (min)	B (min)	C (max)	D (max)	X (max)
1/4	-6	1-3/16-12 UN	0.62	1.25	1.50	3.25	2.47
3/8	-6	1-3/8-12 UNF	0.62	1.44	1.25	3.25	2.47
3/8	-8	1-3/8-12 UNF	0.62	1.44	1.50	3.50	3.00
3/8	-12	1-3/4-12 UN	0.62	1.44	1.75	3.75	3.50
1/2	-12	1-3/4-12 UN	0.62	1.81	2.00	4.00	3.50
1/2	-16	1-3/4-12 UN	0.62	1.81	2.25	4.25	4.81
3/4	-16	2-12 UN	0.62	2.12	2.25	4.50	4.81
1	-16	2-5/16-12 NS	0.75	2.38	2.25	4.75	4.31
1	-20	2-5/16-12 NS	0.75	2.38	2.75	5.25	4.81
1-1/4	-16	2-3/4-12 UN	0.75	2.88	2.25	5.00	4.81
1-1/4	-20	2-3/4-12 UN	0.75	2.88	2.75	5.50	4.50
1-1/4	-24	2-3/4-12 UN	0.75	2.88	3.00	5.25	5.38
1-1/2	-32	3-1/16-12 NS	0.88	3.12	3.50	6.00	5.81
2	-32	3-3/4-12 UN	0.88	3.88	3.50	6.00	5.81

SH 12298

NOTES:
1. Dimensions are in inches.
2. Material may be to MIL-T-1368 or MIL-T-16420.

FIGURE 4. Type I O-ring union, male straight end fitting.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T19:30Z
Check the source to verify that this is the current version before use.



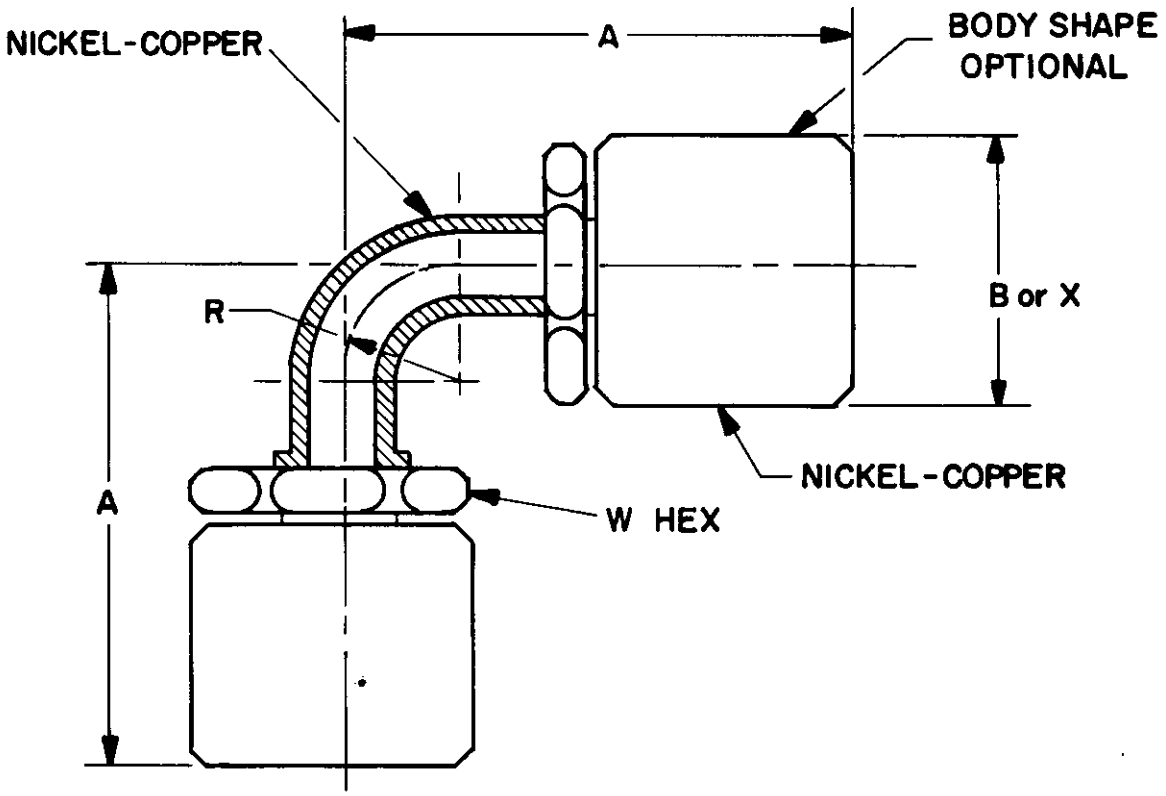
Tube size (nom)	Hose end size (dash no.)	T Thread CL-3A	A (min)	B (max)	C (min)	D (max)	E (max)	R (nom)	X Rad (min)	W Hex (nom)
1/4	-6	1-3/16-12 UN	0.62	1.25	1.50	2.25	3.55	0.50	2.47	0.94
3/8	-8	1-3/8-12 UNF	0.62	1.44	1.50	2.50	3.90	0.75	3.00	1.00
1/2	-12	1-3/4-12 UN	0.62	1.81	2.00	3.25	4.75	1.25	3.30	1.31
3/4	-16	2-12 UN	0.62	2.12	2.25	3.75	5.75	1.50	4.31	1.62
1	-20	2-5/16-12 NS	0.75	2.38	2.75	4.25	6.75	1.75	4.81	2.00
1-1/4	-24	2-3/4-12 UN	0.75	2.88	3.00	5.00	7.00	2.00	5.38	2.00
1-1/2	-32	3-1/16-12 NS	0.88	3.12	3.50	6.00	8.25	2.50	5.81	2.75
2	-32	3-3/4-12 UN	0.88	3.88	3.50	6.00	8.25	2.50	5.81	2.75

SH 12299

NOTES:
1. Dimensions are in inches.
2. Material of elbow may be to MIL-T-1368 or MIL-T-16420.

FIGURE 5. Type IL O-ring union, male 90 degrees end fitting.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:30Z
Check the source to verify that this is the currentversion before use.



Rubber hose (dash no.)	A (max)	B (max)	R (nom)	X (max)	W (Hex) (nom)
-6	3.25	1.50	0.50	2.47	0.94
-8	3.75	1.50	.75	3.00	1.00
-12	4.75	2.25	1.25	3.50	1.31
-16	5.75	2.25	1.50	4.31	1.62
-20	6.75	2.75	1.75	4.81	2.00
-24	7.00	3.00	2.00	5.88	2.25
-32	8.00	3.50	2.50	5.81	2.75

SH 12300

NOTES:
1. Dimensions are in inches.
2. Material of elbow may be to MIL-T-1368 or MIL-T-16420.

FIGURE 6. Type E 90 degree elbow fittings.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:30Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL SEA SYSTEMS COMMAND (SEA 55Z3)
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20362

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:30Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24136/2(SH)	Hose, Synthetic Rubber, Synthetic Fiber Reinforced And End Fittings, Reusable, For Flexible Hose Assemblies

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-19T19:30Z
Check the source to verify that this is the current version before use.